# Practice workbook · Judge whether data support a claim and spot their limits

Health and investigation data where the student decides if evidence supports a claim, why an estimate is uncertain, which result is anomalous, or why data are unreliable or incomplete.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-HEALTH-DATA-CLAIMS

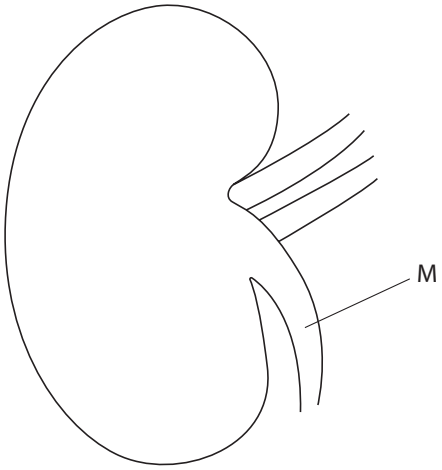
# 1. Question 7 · part c.iii · 4 marks

June 2022 Paper 2R

Worked-example retry · Attempt part c.iii only.

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**7** The diagram shows a kidney.




(a) (i) Which of these is the name of structure M?

**(1)**

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

**(1)**

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

# 1. Question 7 · part c.iii · 4 marks

June 2022 Paper 2R

Worked-example retry · Attempt part c.iii only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

| Substance excreted during 24 hours | High protein diet Mass in g | Low protein diet Mass in g |
|---|---|---|
| total nitrogen compounds | 16.8 | 3.6 |
| urea | 14.7 | 2.2 |
| uric acid | 0.18 | 0.09 |
| ammonia | 0.49 | 0.04 |
| creatinine | 0.58 | 0.60 |

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

**(3)**

percentage difference = ..... %

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# 1. Question 7 · part c.iii · 4 marks

June 2022 Paper 2R

Worked-example retry · Attempt part c.iii only.

(ii) Which is the main nitrogen containing excretory product produced from the breakdown of protein in the diet?

Use the information in the table to explain your answer.

**(2)**

(iii) The breakdown of muscle tissue continues at a constant rate and is not affected by diet.

Discuss whether the information in the table supports this statement.

**(4)**

**(Total for Question 7 = 14 marks)**

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## 1. Question 7 · part c.iii · 4 marks · Working and answer

Working and answer

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## 2. Question 9 · part a.ii · 1 mark

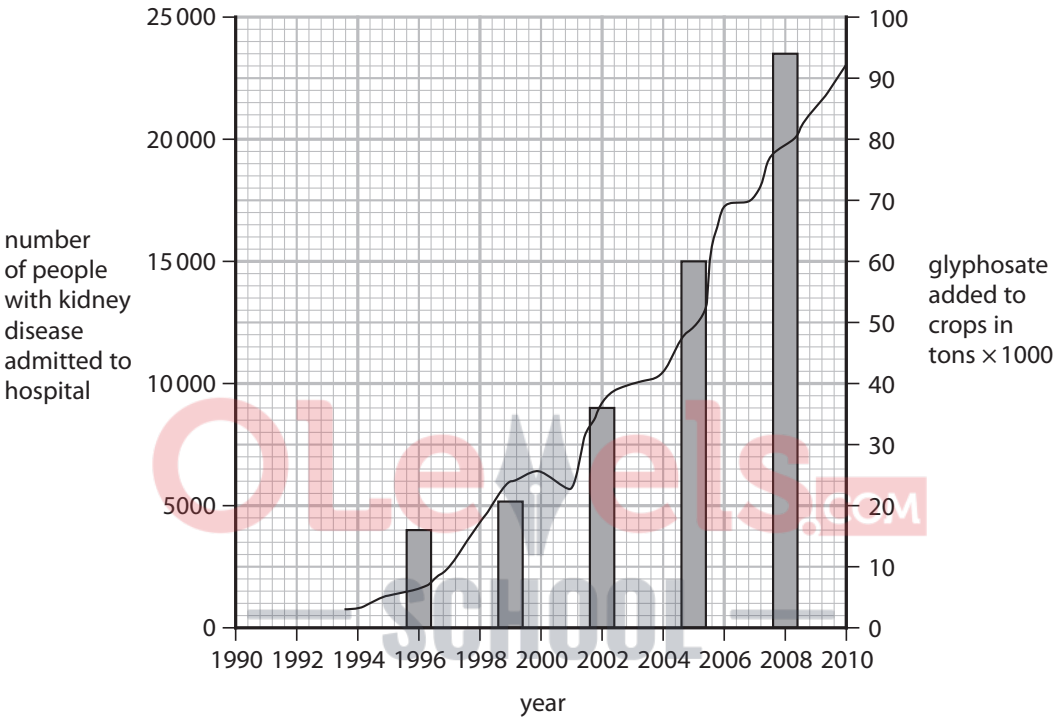
November 2020 Paper 1R

Further practice · Attempt part a.ii only.

**9** (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.




(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

**(3)**

percentage increase = ..... %

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**20**

P 6 2 0 5 2 A 0 2 0 2 4

## 2. Question 9 · part a.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

**(1)**

.....

.....

(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

**(4)**

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## 2. Question 9 · part a.ii · 1 mark · Working and answer

Working and answer

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# 3. Question 7 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

**7** A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 $cm^3$ per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

| **Year** | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|
| **GFR in $cm^3$ per minute** | 80 | 75 | 79 | 79 | 67 | 60 | 50 | 45 | 35 |

(a) (i) Plot a line graph of the data on the grid.

**(4)**



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# 3. Question 7 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

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(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

**(2)**

(iii) A person needs dialysis if the GFR falls to 15 $cm^3$ per minute.

Use the data to estimate when the person will probably need to start dialysis.

**(2)**

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(iv) Discuss the accuracy of this estimate.

**(3)**

## 3. Question 7 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

(b) During the early stages when a person's kidneys are not working correctly, they are advised to control the amount of protein in their diet.

Explain why the protein in their diet should be controlled.

**(3)**

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**(Total for Question 7 = 14 marks)**

## 3. Question 7 · part a.iv · 3 marks · Working and answer

Working and answer

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# 4. Question 6 · part a.ii · 2 marks

June 2019 Paper 1

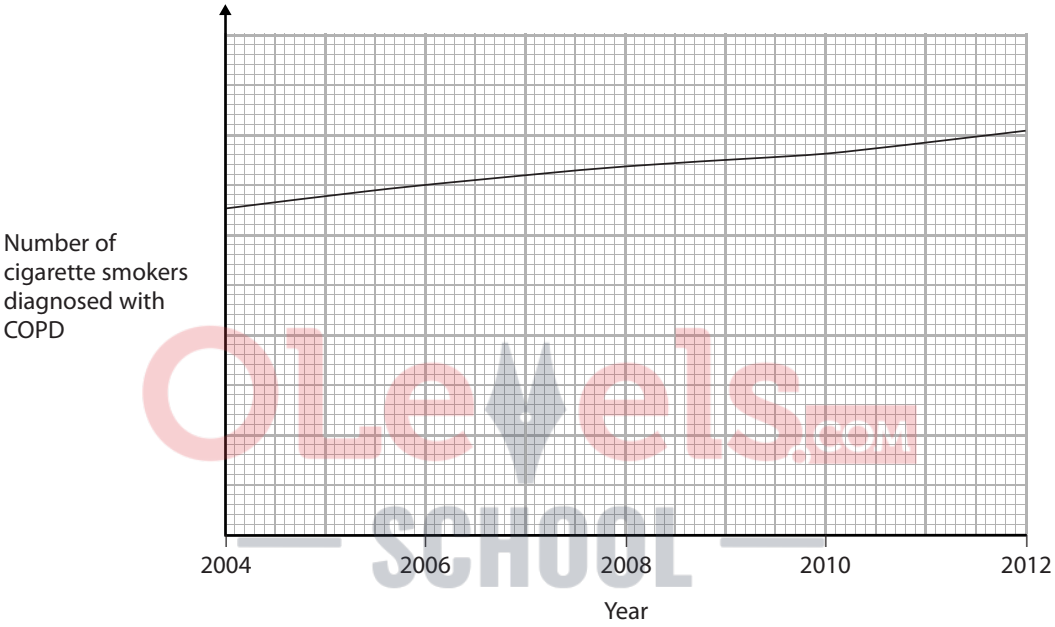
Further practice · Attempt part a.ii only.

**6** (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.




(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

**(1)**

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

**(2)**

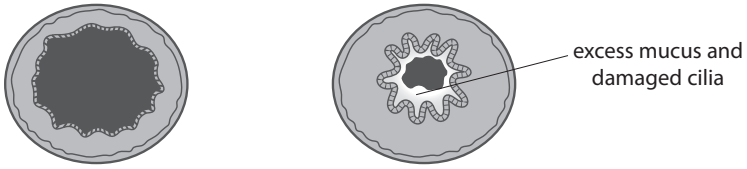
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# 4. Question 6 · part a.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.ii only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.




healthy airway

chronic bronchitis

Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

**(2)**

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## 4. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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# 5. Question 7 · part a.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

**7** The table gives the percentage of overweight people in different age groups for six countries.

| Country | Percentage overweight | | Percentage overweight for each age group | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Males | Females | 18–24 | 25–34 | 35–44 | 45–64 | 65–74 | 75 years and over |
| Croatia | 67.5 | 48.3 | 22.3 | 38.1 | 53.7 | 69.2 | 73.1 | 65.7 |
| Malta | 66.8 | 55.2 | 36.0 | 52.5 | 62.7 | 69.0 | 74.1 | 65.3 |
| Poland | 64.1 | 46.7 | 19.3 | 38.7 | 52.8 | 66.0 | 73.2 | 65.5 |
| Portugal | 57.6 | 50.0 | 23.6 | 35.1 | 49.9 | 63.5 | 69.3 | 71.2 |
| Turkey | 56.4 | 56.3 | 22.4 | 44.2 | 64.9 | 75.4 | 66.2 | 56.5 |
| UK | 60.2 | 51.8 | 29.0 | 47.3 | 56.0 | 64.0 | 62.8 | 52.6 |

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

**(2)**

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ratio = .....

(ii) Give two trends shown in the data.

**(2)**

1 .....

.....

2 .....

.....

(iii) Give two reasons why the data for a country might be unreliable.

**(2)**

1 .....

.....

2 .....

.....

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## 5. Question 7 · part a.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

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(b) (i) Individuals who are overweight or obese have an increased risk of diabetes.

People with diabetes can have high blood glucose levels.

Explain how blood glucose levels are reduced in a person who does not have diabetes.

**(3)**

(ii) Suggest why one symptom of diabetes is frequent urination.

**(2)**

**(Total for Question 7 = 11 marks)**

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## 5. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

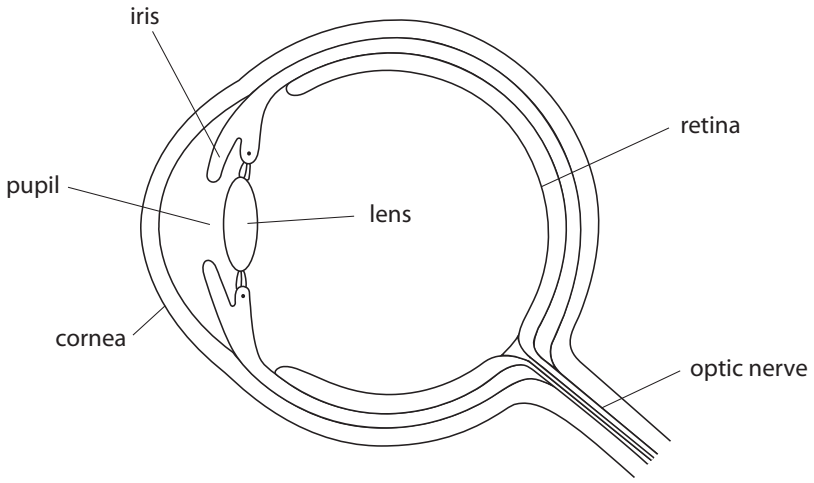
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# 6. Question 3 · part b.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

**3** The human eye contains structures that help to focus light on the retina.




(a) Which structures help to focus light on the retina?

**(1)**

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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# 6. Question 3 · part b.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

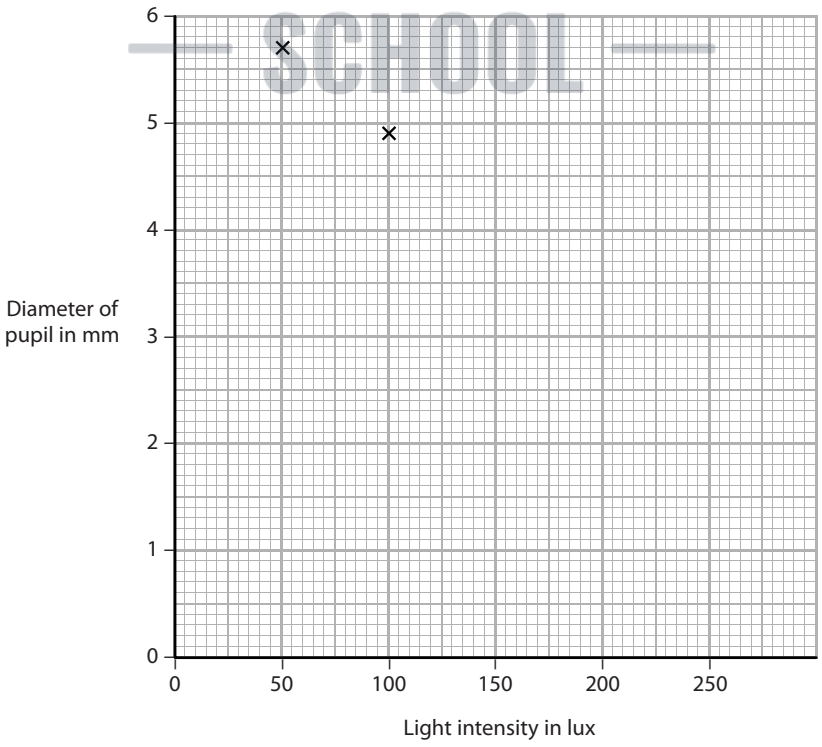
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

| Light intensity in lux | Diameter of pupil in mm |
|---|---|
| 50 | 5.7 |
| 100 | 4.9 |
| 150 | 2.7 |
| 200 | 3.2 |
| 250 | 2.3 |

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)




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## 6. Question 3 · part b.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

(iii) Explain which result is anomalous.

**(2)**

(iv) State the effect of light intensity on the diameter of the pupil.

**(1)**

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

**(2)**

**(Total for Question 3 = 8 marks)**

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## 6. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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# 7. Question 7 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

**7** A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

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(a) (i) Plot a line graph of the data on the grid.

**(4)**



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# 7. Question 7 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

**(2)**

(iii) A person needs dialysis if the GFR falls to 15 $cm^3$ per minute.

Use the data to estimate when the person will probably need to start dialysis.

**(2)**

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(iv) Discuss the accuracy of this estimate.

**(3)**



P 7 2 5 8 5 A 0 1 9 2 0

## 7. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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## 8. Question 6 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

**6** (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

| Region | Total number of people tested | Percentage of colour blind males | Percentage of colour blind females |
|---|---|---|---|
| A | 150 | 10.00 | 1.00 |
| B | 16 000 | 6.60 | 0.40 |
| C | 5000 | 0.80 | 0.01 |

(i) Calculate the number of colour blind males in the test group in region C.

**(3)**

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number of males = .....

(ii) State why the results from region A are not considered to be reliable.

**(1)**

.....

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## 8. Question 6 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

(iii) Justify the statement that colour blindness is a recessive sex-linked condition.

**(5)**

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## 8. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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